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Practical 4

The screenshot displays the MySQL Workbench interface. The left sidebar shows a list of schemas, including 'bookdonation_db', 'bookflow_db', 'capacity_connect_db', 'sakila', 'students_db', 'sys', 'university', and 'world'. The main editor window contains the following SQL script:

```
1 DROP DATABASE IF EXISTS Bookflow_db;
2
3 CREATE DATABASE Bookflow_db;
4
5 USE Bookflow_db;
6
7 CREATE TABLE Books
8 (
9     book_id INT AUTO_INCREMENT PRIMARY KEY,
10    title VARCHAR(200) NOT NULL,
11    author VARCHAR(150) NOT NULL,
12    genre VARCHAR(100),
13    description TEXT,
14    published_year INT
15 );
16
17 INSERT INTO Books
18 (
19     title,
20     author,
21     genre,
22     description,
23     published_year
24 )
25 VALUES
26 (
27     'The Martian',
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
28	21:31:43	CREATE DATABASE Bookflow_db	1 row(s) affected	0.032 sec
29	21:31:46	USE Bookflow_db	0 row(s) affected	0.000 sec
30	21:31:50	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected	0.110 sec
31	21:31:54	INSERT INTO Books (title, author, genre, description, published_year)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.015 sec
32	21:31:59	VALUES ('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive alone on M...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
33	21:32:15	VALUES ('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive alone on M...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookdonation_db
- bookflow_db
- capacity_connect_db
- sakila
- students_db
- sys
- university
- world

Administration Schemas

Information

No object selected

Query 1 SQL File 4* SQL File 5* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*

Limit to 1000 rows

```
93 'A fantastic underwater adventure aboard a mysterious submarine.',
94 1870
95 );
96
97 ALTER TABLE Books
98 ADD COLUMN embedding JSON;
99
100 UPDATE Books
101 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
102 WHERE book_id = 1;
103
104 UPDATE Books
105 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
106 WHERE book_id = 2;
107
108 UPDATE Books
109 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
110 WHERE book_id = 3;
111
112 UPDATE Books
113 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
114 WHERE book_id = 4;
115
116 UPDATE Books
117 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
118 WHERE book_id = 5;
119
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

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31	21:31:54	INSERT INTO Books (title, author, genre, description, published_year)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''An astronaut struggles to survive alone on M...' at line 1	0.015 sec
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Object Info Session

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Limit to 1000 rows

```
121 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
122 WHERE book_id = 6;
123
124 UPDATE Books
125 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
126 WHERE book_id = 7;
127
128 UPDATE Books
129 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
130 WHERE book_id = 8;
131
132 UPDATE Books
133 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
134 WHERE book_id = 9;
135
136 UPDATE Books
137 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
138 WHERE book_id = 10;
139
140 SELECT
141 book_id,
142 title,
143 author,
144 genre,
145 published_year,
146 embedding
147 FROM Books
```

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Limit to 1000 rows

```
225 ),
226 2
227 )
228 +
229 POW(
230 CAST(
231 JSON_EXTRACT(embedding, '$[2]')
232 AS DECIMAL(10,6)
233 ),
234 2
235 )
236 )
237 *
238 Sqrt(
239 POW(0.90, 2)
240 +
241 POW(0.80, 2)
242 +
243 POW(0.20, 2)
244 )
245 ),
246 4
247 ) AS cosine_similarity
248 FROM Books
249 WHERE embedding IS NOT NULL
250 ORDER BY cosine_similarity DESC
251 LIMIT 3;
```

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31	21:31:54	INSERT INTO Books (title, author, genre, description, published_year)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version [4.0.15.1]	0.015 sec
32	21:31:59	VALUES ('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive alone on Mars', 2015)	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version [4.0.15.1]	0.000 sec
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```
1 CREATE DATABASE bookflows_db;
2 USE bookflows_db;
3
4 CREATE TABLE Books (
5 book_id INT PRIMARY KEY,
6 title VARCHAR(100) NOT NULL,
7 isbn VARCHAR(20) UNIQUE,
8 published_year INT,
9 CHECK (published_year <= 2027)
10 );
11
12 CREATE TABLE Members (
13 member_id INT PRIMARY KEY,
14 full_name VARCHAR(100) NOT NULL,
15 email VARCHAR(100) NOT NULL UNIQUE
16 );
17
18 INSERT INTO Books
19 (book_id, title, isbn, published_year)
20 VALUES
21 (1, 'The Alchemist', '9780061122415', 1988),
22 (2, 'Atomic Habits', '9780735211292', 2018),
23 (3, 'Rich Dad Poor Dad', '9781612681139', 1997);
24
25 INSERT INTO Members
26 (member_id, full_name, email)
27 VALUES
```

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